

Mobile AR Geometry and Linear Algebra Tutor

Xiaolong Ge

School of Computer Science
University of Auckland, Auckland, New Zealand
xge666@aucklanduni.ac.nz

ABSTRACT

This paper aims to develop an innovative educational software platform that leverages modern, interactive teaching tools to enhance student understanding and engagement. The core of our approach involves the integration of tangible augmented reality (AR) and multimodal learning strategies, thereby fostering multi-modal learning. We aim to develop instructional software for linear algebra and geometry using AR through Unity3D on mobile platforms, preferably supports both iOS and Android. Our approach uses AR as a visualisation approach, enabling students to interact with real-world objects in real time to visualize mathematical operations such as vector addition and matrix transformations. The app enables users to quickly and accurately match the corresponding mathematical functions, which enhances students' understanding of these key concepts involved. The results show that students utilizing the app consistently attained high scores, reaching up to 85.71% correctness, in contrast to 35.11% on average for those employing traditional slides. In addition, the app users enhanced their scores by 57.7% overall compared to those utilizing slides, demonstrating the app's utility for the users. Finally, users who were able to accurately and quickly match the mathematical functions performed better in terms of conceptual understanding.

Author Keywords

Augmented Reality; Unity; Mobile Application; Linear Algebra; Geometry

INTRODUCTION

Geometry and linear algebra are foundational subjects for secondary education, serving as essential building blocks for a wide array of university-level courses and disciplines. However, these subjects often present significant challenges to students due to their inherently abstract nature and the need for advanced spatial reasoning skills. Concepts in geometry and linear algebra can be difficult to visualize and comprehend using traditional instructional methods, such as textbooks or static diagrams. In New Zealand, while schools introduce basic geometry in primary education, advanced topics such as

vector and matrix calculus are no longer part of the standard curriculum unless students opt for specialized mathematics courses. This decline in instructional focus has impeded students' ability to develop a solid understanding of these crucial subjects [1, 2]. This paper aims to address these challenges by developing an innovative educational software platform that leverages modern, interactive teaching tools to enhance student understanding and engagement.

The core of our approach involves the integration of tangible augmented reality (AR) and multi-modal learning strategies. Augmented reality provides a dynamic medium through which abstract mathematical concepts can be represented in a three-dimensional format, allowing learners to visualize and manipulate these concepts within a real-world context. Tangible AR, specifically, facilitates learning by enabling students to use physical objects to interact with virtual content, thus making abstract mathematics more intuitive. For example, using physical objects such as pens to represent vectors allows students to more naturally engage with and understand operations like vector addition, subtraction, dot products, and cross products. This multi-modal approach—incorporating tactile, visual, and spatial learning—has significant potential for enhancing comprehension and retention of highly abstract concepts such as those found in geometry and linear algebra [15].

In addition to the advantages of AR, accessibility considerations are crucial. Although mobile devices are common, they are not universally available to all students. Given that most schools require students to use devices such as Chromebooks or tablets, our platform will be developed to ensure compatibility with these devices to maximize accessibility and inclusion.

It is important to emphasize that this research focuses on the specific pedagogical effects of tangible AR, rather than incorporating gamification elements. While gamification can enhance motivation, it also has the potential to confound our investigation into the educational impact of tangible AR itself. Our primary objective is to determine how tangible AR can facilitate an intuitive grasp of complex concepts through direct interaction with real-world objects.

RELATED WORK

The 1994/1995 Third International Mathematics and Science Study (TIMSS) found that mathematics education in New Zealand fell below the international average across key areas, including number operations, measurement, and algebra [1, 2, 3]. However, these findings primarily relate to general numeracy, which differs in scope from the specific challenges posed by geometry and linear algebra. To effectively address

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, to republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CompSci 715, Semester 2, 2020, Auckland, New Zealand

© 2020 Copyright held by the owner/author(s). Publication rights licensed to ACM. ISBN 123-4567-24-567/08/06...\$00.00

DOI: http://dx.doi.org/10.475/123_4

the learning difficulties inherent in these areas, targeted interventions are necessary.

There is a substantial body of research on the use of AR in the teaching of geometry and algebra. For instance, Cahyono et al. [4] employed pre-set 3D models to assist students in visualizing geometric concepts, allowing them to interact with the models to better understand related formulas. Kaufmann et al. [5] introduced Construct3D, an AR-based educational tool that utilized head-mounted displays to enable students to manipulate geometric constructions in a virtual space. Their study demonstrated significant improvements in students' spatial skills, particularly with respect to problem-solving, underscoring the potential of AR as a means of enhancing mathematical comprehension.

In the domain of linear algebra, Singh et al. [6] developed a Unity3D-based system that facilitated student interaction with matrix elements through real-time modeling, fostering a deeper understanding of transformations such as translation, rotation, and scaling. Similarly, Cerqueira et al. [7] examined how AR games could contribute to education by creating an interactive soccer game called FootMath, which made learning algebraic functions more engaging and contextually meaningful.

Recent advancements in tangible AR have highlighted its potential for enhancing educational outcomes by combining real-world object manipulation with virtual overlays, thereby fostering multi-modal learning. Smith et al. [9] demonstrated that tangible AR could significantly improve learners' spatial reasoning abilities by providing immediate, context-sensitive feedback through the manipulation of physical objects. This approach bridges the gap between abstract theoretical constructs and concrete experiential understanding, which is particularly beneficial for subjects like geometry and linear algebra.

Monteiro et al. [15] introduced "Teachable Reality," an AR prototyping tool that empowers users to create interactive tangible AR experiences by leveraging everyday objects as input devices. By utilizing interactive machine teaching, the tool enables real-time creation of tangible interactions without requiring programming expertise. Their findings suggest that this approach significantly reduces barriers to entry for AR prototyping, offering a high degree of flexibility and customization that complements existing marker-based methods. Such flexibility is highly relevant for our research, as it allows learners to actively engage with and explore complex mathematical concepts through direct, hands-on interaction.

The study by Zhang et al. [8] demonstrated how AR can be used effectively for visualizing complex geometric transformations, providing empirical evidence of improved spatial understanding and engagement among students. Similarly, Smith et al. [9] explored the integration of AR in mathematics education, focusing on its effects on student motivation and cognitive load. Their research indicates that while AR can enhance engagement, careful design is required to prevent cognitive overload.

In another study, Johnson et al. [10] presented an intelligent AR system for learning geometry that adapts to individual student needs through real-time feedback. This adaptive system provides targeted assistance based on user interactions, thus supporting personalized learning pathways. The work by Chen et al. [11] further explored the use of AR in the context of algebraic learning, emphasizing the role of interactive visualization in reducing misconceptions and enhancing conceptual understanding. Lastly, the research by Liu et al. [12] highlighted the importance of using AR to create collaborative learning experiences, which can foster peer learning and improve problem-solving skills in mathematics.

VanDieren et al. [16] explored student understanding of vectors and cross products through a series of visualization tasks, emphasizing the challenges students face with these operations. The study identified key areas of conceptual confusion, including differentiating between dot products and cross products, and interpreting the geometric meaning of these operations. The authors argue that improving students' visualization skills using tools like CalcPlot3D can enhance their comprehension of vector operations in a multivariable calculus context. This finding aligns with the goals of our research in using tangible AR to support students' understanding of complex mathematical concepts by providing visual and interactive representations.

In summary, current applications of AR in education often rely on pre-built 3D models and platforms to facilitate student engagement with abstract concepts. Our approach seeks to advance this paradigm by incorporating tangible AR, enabling students to interact with real-world objects in real time to visualize mathematical operations such as vector addition and matrix transformations. This method aims to make highly abstract mathematical concepts more accessible, engaging, and comprehensible for secondary school students, thereby enhancing their foundational knowledge and preparing them for more advanced study.

RESEARCH PROBLEM

In this paper, we will focus on these research questions:

Research Problem 1

What are the threshold concepts of learning introductory geometry and linear algebra?

Research Problem 2

What are the benefits of using tangible AR for improving learning of introductory geometry and linear algebra?

Research Problem 3

What are suitable learning tasks for a tangible AR learning tool for introductory geometry and linear algebra?

Research Problem 4

What formative feedback can be automatically produced for these learning tasks from the users' interactions with the AR tool?

Research Problem 5

What are its effects of this feedback on students' understanding?

INTENDED METHODOLOGY

We intend to develop instructional software for linear algebra and geometry using augmented reality through Unity3D on mobile platforms, preferably supports both IOS and Android. This will involve development on mobile devices; related toolkits in the operating systems may help. The software will explore and utilise gamification techniques, including but not limited to setting up narrative guides or integral breakout levels, to engage students in learning relevant mathematical concepts and testing results. We hope to achieve two goals in the software: Firstly, pre-build examples of geometric concepts or linear algebra (e.g., matrix calculations or common 3D geometric shapes) in the software so that the users can learn the mathematical concepts by adjusting the parameters and observing the changes. The second is to add a scanning function; users can scan some flat geometries and generate real-time 3D models within the software in order to understand those uncommon or non-prebuilt special cases to deepen the understanding of the relevant mathematical concepts.

TECHNOLOGY EVALUATION

Progress is expected to be advanced on a weekly basis, with all elements completed by October 14th. Need to learn how to develop virtual reality applications using Unity3D, mobile toolkits and development-related knowledge. Also to further explore forms of gamification and how to build 3D models in real time through scanning.

DESIGN

Based on our research questions, the main focus of our research lies in identifying mathematical concepts that are suitable to be categorised at the beginner's level of geometry and linear algebra, and setting appropriate task objectives based on these concepts. The second is developing AR software that implements enough functionality to accomplish these goals. In order to accomplish this, we also need to make/select suitable 3D objects as user-accessible targets that the user will manipulate to complete the tasks. Then, the participants are called in for a user study to find out the feedback on the possible improvement of learning these mathematical concepts using AR compared to traditional methods. The Figure 1 below shows the overall architectural flow of our study.

The figure 2 below shows the workflow of our user study.

Next, We'll lay out the design ideas for each section in modules.

Requirements

To be able to address our second research question, we designed an AR app to collect user research data. Its main purpose is to allow users to visually experience the basic concepts of geometry and linear algebra that we have set up, and learn the steps to complete the relevant calculations and demonstrations guided by hints to help establish a proper understanding of the concepts involved. Inspired by Singh, G. et al. [6], the following is a list of specific requirements that our project aims to achieve after research and summarisation.

1. The App is able to track objects in real-time
2. Objects are marked/made for easy tracking

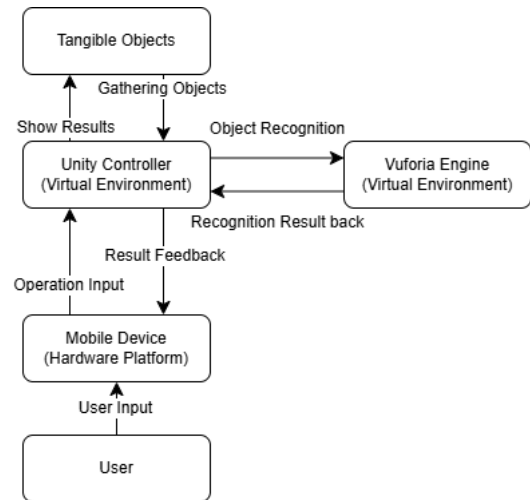


Figure 1. Overall Architectural Flow

3. Variety of different algebra/geometry exercises
4. Mathematical/graphical information in overlay
5. Formative feedback displayed to help student
6. Hints to guide the users to use the App

Hardware Platform

Our app is designed to be a teaching tool that meets the needs of anytime, anywhere use, thus making mobile devices the main target operating platform. In addition, as tablets and laptops are often too large to be hand-held and rely on external stands or platforms when used as stationary devices, we decided to use mobile phones as the target hardware for our app.

As we all know nowadays, the mobile phone platform is mainly divided into two camps, which are the Android that belongs to Google and iOS platform which is limited to Apple only. For Android, there have many other open-source release versions distributed by third-party manufacturers. Because so many different systems exist, it makes adapting to each of them very difficult. Therefore, it becomes critical to choose a suitable platform as the main target for development. After some research and study, we have found that there do have some significant advantages when developing AR projects on iOS platform as compared to Android, as follows:

Hardware consistency and optimisation

The relatively unified hardware ecosystem of iOS devices allows AR developers to take full advantage of the hardware capabilities of each generation of devices. For example, the LiDAR (laser radar) sensors integrated in the iPhone 12 and later devices provide highly accurate depth perception, which is particularly important in environment modelling and object detection. Unlike iOS, Android devices come from many different manufacturers and have a wide range of hardware combinations, with processing power and camera specifications varying greatly from device to device. AR apps need to be adapted to a wide range of hardware, which may result in

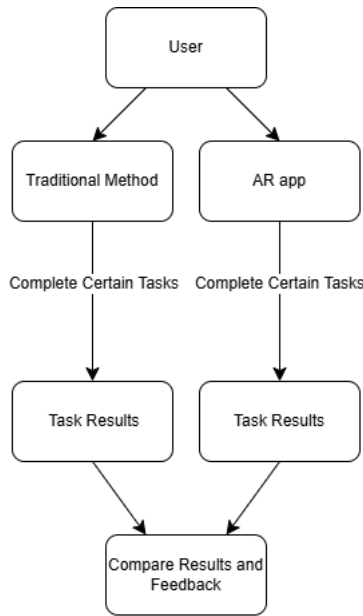


Figure 2. User Study Workflow

the apps not being able to take full advantage of the app’s capabilities on certain devices, and a more complex development and debugging process.

According to the study, LiDAR not only improves the smoothness of the AR experience, but also significantly improves performance in low-light environments, whereas Android devices are more pronounced in terms of hardware fragmentation, leading to large differences in performance and AR experience between devices. [13, 14]

Using the iPhone’s LiDAR feature, developers can design AR apps for accurate indoor navigation or 3D environment modelling, which has great potential in areas such as architectural design and interiors, whereas there are only a handful of high-end models of Android devices that are capable of similar functionality. (e.g. high-end Samsung or Pixel phones)

The Power of ARKit

ARKit, launched by Apple, is an augmented reality development framework optimised for iOS devices, offering a wide range of efficient features and tools. For example, ARKit supports facial recognition-based AR, accurate plane detection, light estimation, multiplayer AR experiences, and other features with a high level of integration that developers can easily use to create complex AR applications. Google developed ARCore for Android, but due to the diversity of Android devices, ARCore’s performance, accuracy, responsiveness, and stability are slightly inferior compared to ARKit, especially when it comes to complex motion capture and light estimation. These features play an important role in enhancing user interactivity and experience, making the development of mobile AR applications using ARKit a unique advantage.

On iOS, some AR games take advantage of ARKit’s multiplayer interaction feature, which allows multiple users to participate in the game at the same time, sharing AR objects

and interacting with them in real time, while the implementation on Android is relatively complex and inconsistently supported. [13]

Measurement and accuracy advantages

A paper suggests that ARKit performs better at distance measurement, especially in application scenarios that require high accuracy. For example, AR apps for iOS can provide more accurate physical measurements with ARKit, while Android’s ARCore has larger errors on some devices. This makes iOS more suitable for architectural design, engineering measurements, and other AR applications that require high accuracy. [13]

Ecosystem of Development Tools

XCode, the development tool for iOS, is deeply integrated with Apple’s ecosystem and provides a very smooth development experience, especially in AR projects. XCode’s built-in AR debugging tools and simulators help developers develop and test faster. In addition, the Swift language and Metal graphics API enable more efficient control of hardware resources on iOS devices, providing superior performance for AR applications. While Android Studio is powerful enough for developing regular Android apps, in AR projects, Android Studio’s AR debugging tools and simulators are not as mature as XCode, and the debugging experience may vary significantly from device to device. For example, when developers use XCode for AR project development, they can debug the rendering effect of the AR scene directly in the simulator and can test ARKit’s plane detection and light estimation functions in real time. In Android Studio, although you can use virtual devices for some AR debugging, but many high-end features still need to be tested on real devices.

Based on the above advantages, we decided to target iOS as the development platform so that we could maximise the advantages of the hardware.

Virtual Environment

The Virtual Environment is mainly consisted of two parts, we use Unity3D as the development platform as well as the controller of the APP, and the object recognition is powered by the Vuforia Engine SDK.

Controller: Unity3D

We decided to use Unity3D as the development platform and controller is based on the following considerations.

1. Cross-platform compatibility

One of the biggest advantages of Unity is its strong cross-platform compatibility. Developers can easily port their apps to iOS, Android, and other devices such as HoloLens or Magic Leap in one development. This compatibility not only saves development time, but also reduces maintenance costs across different platforms. This makes it possible to port what we have tested on Windows to iOS in a very good way.

2. Strong development ecosystem and plugin support

Unity provides rich support for plug-ins and tools, such as Vuforia and AR Foundation. Vuforia allows developers to support iOS’s ARKit and adapt to Android’s ARCore to a

certain extent through the same codebase, thus simplifying the development of cross-platform AR applications. In addition, Unity's Asset Store provides a wealth of 3D models, textures, sound effects, and other resources to help developers quickly build complex AR scenes.

3. Flexible Scripting Capabilities

Unity uses C# as its main programming language and provides highly flexible scripting capabilities. Developers can easily implement complex logic and interactions and develop them modularly through Unity's component system. This allows developers to create not only simple AR experiences, but also highly interactive applications.

Object Recognition & Tracking: Vuforia Engine

Before deciding to use Vuforia as our target recognition component, we explored the effectiveness and limitations of many other different target recognition methods.

AR Foundation + ARCore

ARCore is an Augmented Reality component developed by Google, and in Unity, it relies on the AR Foundation, which allows for cross-platform use of the Augmented Reality platform in Unity but does not implement any AR functionality itself. Therefore, target recognition with this solution depends entirely on the functionality of ARCore itself. However, ARCore does not support tracking of 3D objects and, therefore, is not suitable for our project. As the Figure 3 shows.

	ARCore	ARKit	Magic Leap	HoloLens
Device tracking	✓	✓	✓	✓
Plane tracking	✓	✓	✓	
Point clouds	✓	✓		
Anchors	✓	✓	✓	✓
Light estimation	✓	✓		
Environment probes	✓	✓		
Face tracking	✓	✓		
2D Image tracking	✓	✓	✓	
3D Object tracking		✓		
Meshing		✓	✓	✓
2D & 3D body tracking		✓		
Collaborative participants		✓		
Human segmentation		✓		
Raycast	✓	✓	✓	
Pass-through video	✓	✓		
Session management	✓	✓	✓	✓
Occlusion	✓	✓		

Figure 3. ARCore Support Overview and Comparison

ARKit

Apple's ARKit supports 3D object tracking, but its tracking principle is based on point clouds. Our testing found that it has very poor tracking performance for small volume objects like pens and boxes that we intend to use as tangible objects, so it's not really suitable for that either.

OpenCV

OpenCV allows developers to build target recognition algorithms from scratch with maximum flexibility and scalability. It has 3 ways to use it in Unity3D.

1. Use an off-the-shelf Package, but need to pay for it. (A free version only supports Open CV 3, but many people report that it cannot function well on Unity 2022 and also lacks some of the tracking API)

2. Since OpenCV is written in C++ originally, it needs to be called as an external function in Unity using C#, which is complicated.

3. using the Python version of OpenCV to transfer the data to python for processing via UDP or other protocols, which makes the results significantly slower than C++.

There are 11 methods of target tracking in OpenCV, and we have tested each of them, some of the algorithms perform very well, as shown in Table 1.

Algorithm	Average FPS	Comment
MIL	5	bad result
CSRT	18	bad result
KCF	52	normal result, worse than MEDIANFLOW
BOOSTING	23	bad result
MEDIANFLOW	100	normal result, just worse than MOSSE
TLD	11	bad result
MOSSE	700	best result, but unstable
GOTURN	20	bad result, 300MB big model
NANO	100	good result, 2MB model
VIT	90	good result, 1MB model, a little worse than NANO
DASIAMRPN	30	bad result, 154MB big model, obvious errors

Table 1. Performance Comparison of Different Tracking Models

But unfortunately, all the target tracking methods are based on 2D, and there is no built-in functionality for 3D objects that need to be rotated and moved. This requires us to complete the steps of 3D reconstruction and target recognition by ourselves, which is more difficult.

Vuforia

Vuforia Engine can track images, objects and environment, but its main focus is on images. For 3D objects, it can realize it by tracking model targets, cylinder targets, or multi-targets.

1. Model Targets

This method requires the developer to build a specific model of the object to be tracked through modelling software such as CAD or by using LiDAR scans before it can be used. It has very good tracking performance for specific objects such as industrial parts. But it is not good at tracking small objects, such as pens, and at the same time it is very demanding on the model and performs very poorly if the model does not have enough distinctive features (e.g., symmetrical objects), which is fatal to our vision of wanting to use common objects as tangible objects.

2. Cylinder Targets

The implementation of this method is based on image recognition, users can scan the markers of uploaded cylindrical objects (e.g. Coke labels) to create a cylinder object, to achieve the tracking of the corresponding object, and support rotation and movement.

3. Multi-Targets

Same principle as Cylinder Targets, but multi-targets target box-type objects. It requires uploading all 6 sides of the box to create a Multi-Target, which is a little more stable in terms

of rotation than a Cylinder Target since there are image data for each side.

Based on the comprehensive consideration of the advantages and disadvantages of each method mentioned above, we finally decided to use Vuforia for target recognition and tracking and establish Cylinder Target and Multi-Target as tangible objects.

3D Assets

The 3D Assets we used are basically done by using the components that come with Unity3D itself. For the vector modeling that needs to be done using cones, since Unity3D does not come with this component, we used the Cone Mesh package, which is freely available in the Unity Store (developed by Kevin Castejon) to do it.

Linear Algebra/Geometry Concepts

Based on Research Question 1, we defined the threshold concepts of Linear Algebra and Geometry. We think they are crucial for the entry-level level of learning these subjects as follows.

1. Area of parallelogram created by vectors
2. Angle and distance calculation of vectors
3. Dot product calculation
4. Cross product calculation
5. Angle projection of vectors
6. Vector addition and subtraction
7. Volume of parallelepiped with dot and cross products

Assessment

The research will be assessed by setting up a questionnaire and asking the subjects to complete specific questions. We will divide the subjects into two groups: one group will complete the questions by reading slides and doing calculations on paper and pencil (the traditional way), and the other group will complete the questions by using the app. The results will be quantitatively analyzed by recording videos of the behavior and the total time spent by both groups to complete the questions. Conclusions will be drawn from the feedback from the questionnaire.

IMPLEMENTATION

Our app was developed on a Windows operating system using C# and the Unity3D 2022.3.40f1 game engine. Rider was the main IDE we used to write the code and GitHub was used to synchronize and host the code, ensuring that the group could collaborate.

First, we took pictures of the toothpaste box and soda bottle stickers to capture their image information and uploaded them to Vuforia's cloud backend to create multi-targets and cylinder targets as our tangible objects, imported them into Unity, and set up the Vuforia SDK. After importing them into Unity and setting up the Vuforia SDK, Vuforia will be able to track these targets and map the coordinates to the world coordinates of



Figure 4. Sample image of object tracking being performed with Vuforia

the Unity scene on its own without any additional work, as shown in Figure 4.

Then we convert the previously defined basic concepts into functions and set up a Unity user interface so that the user can choose to execute the desired functions, as shown in Figure 5.



Figure 5. Graph showing the UI interface of the APP

The execution process is as follows: after detecting the tangible objects, the app will attach a vector model to the object model. The length and axes of the vector model are predefined. The vector model follows the tracking object on the fly, which allows us to perform subsequent calculations or functions based on the vector model. All calculations are basically based on Unity's own methods under the Vector3 construct, and we have set up several public variables to make it easier to record the information needed to synchronize between functions. We also drew several prefabs (including the vector model mentioned earlier) to help us implement the functionality, such as a flat prefab to show the parallelogram function and a dashed prefab to assist in showing the projection of the vectors, as shown in Figure 6.

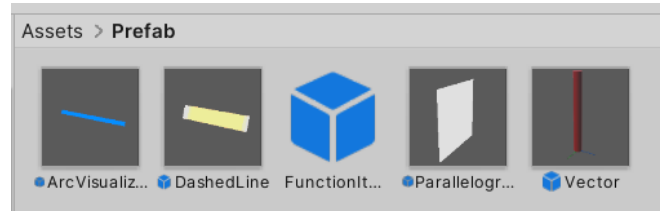


Figure 6. Schematic diagram of the prefabs produced

Specifically, since Unity does not come with the ability to draw the corners of a pinch, we built our own algorithm to draw the corners of the pinch, which is shown in this diagram, as shown in Figure 7.

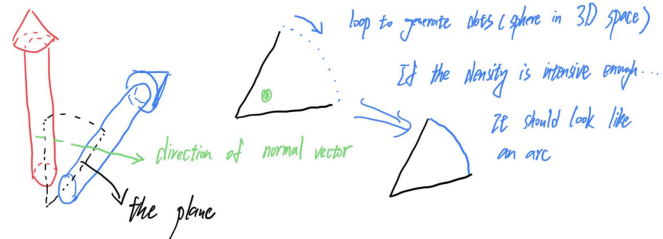


Figure 7. Algorithm schema for visualizing the angle

Finally, the target platform for our deployment is the iPhone, so we need to develop on Unity first, and then export it to XCode, which will convert the program binary into an iOS-ready version, package it, and deploy it to the iPhone. The final program will use the iPhone's rear camera to read the image.

EVALUATION

In order to obtain the effectiveness of our app and answer the research question, we conducted user study using both multiple-choice and open-ended tests and user interviews.

Overview

Participants were divided into two groups. All participants first completed Test 1, and then were assigned to one of the two groups. Group 1 used the app to answer Test 2, followed by answering Test 3 with the aid of a traditional teaching plan (referred to teaching plan). Group 2 first reviewed the teaching plan to answer Test 2, then used the app to answer Test 3. Each segment—Test 1, Test 2, Test 3, reviewing the teaching plan, and using the app—was constrained to ten minutes.

Data Collection Method

The data we collected consisted of two main parts, including responses to open-ended tests and the results of the questionnaire.

In open-ended tests, we designed three test sections, each test contains 3 or 6 linear algebra questions. These questions are designed to measure the test taker's mastery of linear algebra and include both computational and conceptual questions. As 8, 9, 10 below shows.

Test 1

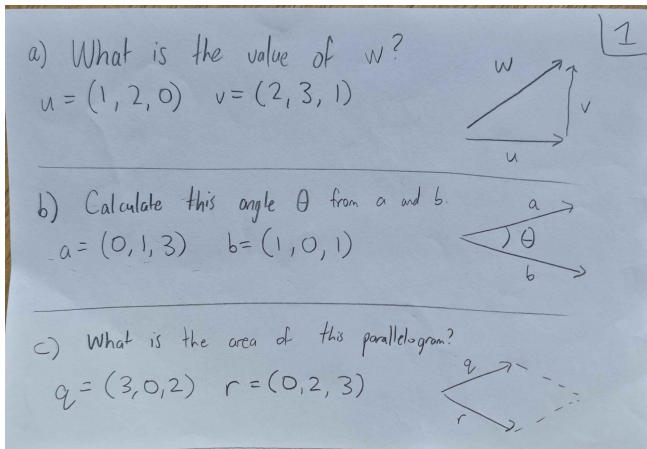


Figure 8. Graph of Test 1 questions

(a): vector operations, involving synthesis and decomposition of vectors.

(b): Calculating angles between vectors, involving dot products of vectors.

(c): Finding the area of a parallelogram, involving the cross product of vectors.

Test 2

(a): Determine the correct graph of the cross product of vectors, involving the geometric representation of the cross product of vectors.

(b): Compare the sizes of the cross products, involving the modulus-angle relationship of the cross products.

(c): calculate the distance from a point to a line, involving vector projections.

(d): calculate the volume of a parallel hexagon, involving mixed products of vectors.

(e): calculate the distance from a point to a plane, involving normal vectors and projections of vectors.

(f): Quiz on dot product involving properties of dot product.

(g): Quiz on Fork Product involving properties of Fork Product.

Test 3

(a): Calculate the cross product of vectors and plot the resultant vector in a coordinate system.

(b): Projecting one vector onto another involves vector projection.

(c): Ask the condition for the maximum value of the dot product, involving the dot product of vectors in relation to angles.

After subjects wrote down as many questions as possible on paper, we collected the results of the subjects' responses to these questions, drafts, and thought processes, scored them according to our designed scoring rubrics, then manually filled

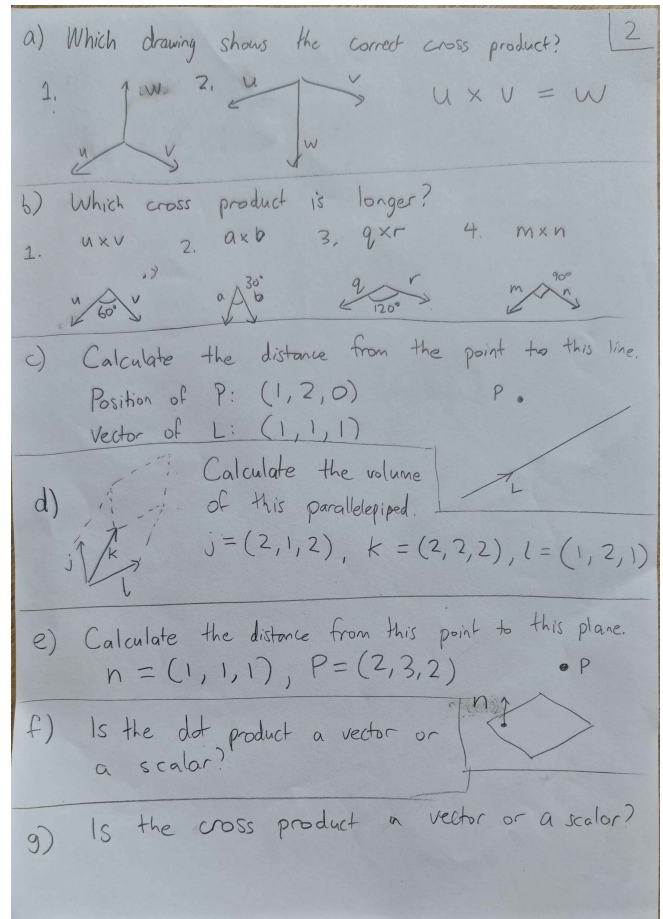


Figure 9. Graph of Test 2 questions

in the spreadsheet and analyzed the data (analyses are elaborated later). Some of the results are shown in the figure, which displays the mathematical problems corresponding to the question numbers (a-f) and the subjects' scores as the 11 shows:

By comparing the differences in scores between different groups of users, the difference in the ability to understand linear algebra between using the app and not using the app can be determined, thus answering our research question 2.

In addition to the open-ended responses, we designed the questionnaire, which was divided into two parts and was asked to be administered at the beginning and at the end of the test. The the questionnaire was distributed through Google Forms and saved in Excel format.

The first part of the questionnaire was mainly based on single-choice questions with our main intention of understanding the user's basic linear algebra level, e.g:

Have you taken any courses or studied topics related to vector operations?

How much experience do you have with 3D first-person perspective games such as Minecraft or Doom?

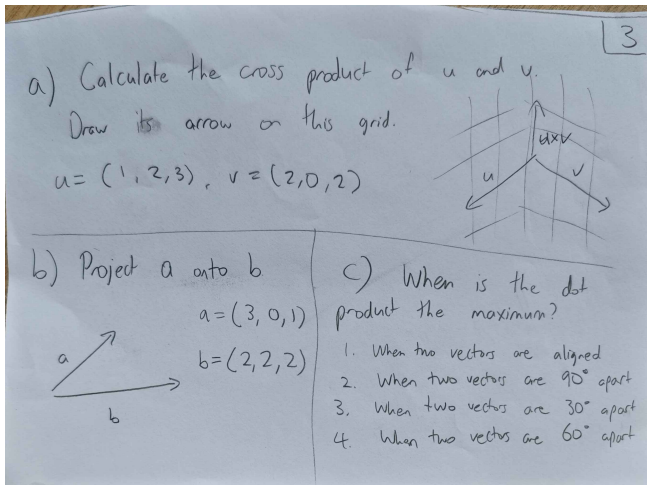


Figure 10. Graph of Test 3 questions

			user 1	user 2	user 3	user 4	user 5
test1	Vector Addition	a	2	6	6	6	6
test1	Dot product	b	0	6	6	2	6
test1	Dot product	c	0	6	6	4	0
test2	cross product	a	0	0	6	0	6
test2	cross product	b	0	6	6	0	6
test2	distance	c	0	0	4	0	2
test2	distance/cross product	d	2	6	6	0	2
test2	distance	e	0	0	0	0	0
test2	Dot product	f	6	6	6	0	6
test2	cross product	g	6	6	6	0	6
test3	cross product	a	4	0	2	0	6
test3	project	b	0	2	4	0	4
test3	dot product	c	6	0	6	0	6

Figure 11. Sample graph of the data collected for 5 users' answers

The second part of the questionnaire was asked to be completed at the end and consisted of multiple-choice questions that required the subjects to score, for example:

How much did the app help you understand the following concepts? (Vector addition and subtraction / Dot product / Cross product / Projection / Vector magnitude)

Which tool did you find more helpful - the lecture slides or the app? (Lecture slides / The app / Unsure)

By collecting the subjects' perceived improvement in their understanding of each piece of linear algebra and comparing this data with the results of the open-ended test. We can know how much the subjects improved their knowledge of each block of linear algebra separately. From this, we can know how effective the subjects were in using the app, and which pieces of linear algebra were learned most effectively using our app, thus answering our third research question.

Data Analysing Method

Test Answer

We scored the answers on the answer sheet, by Score the corresponding linear algebra concepts on the answer sheet and determining the knowledge level of the relevant concepts by scoring each part of the concept in percentage. The specific scoring rubric is shown below.

Each question was worth: 6

Answered the question perfectly: 6

Recognized the formula but miscalculated: 4

Used the formula incorrectly or miscalculated more: 2

Did not answer the question: 0

Then, we will put each question's data and each user's data in a file in a one-to-one correspondence. Then, we will plot a line chart for each user's score, categorized by test, to clearly compare their scores and analyze them. Then, we will separate users into groups based on which group they belong to, the group that used the app first, or the group that used the teaching materials first. We will calculate the average score of each question for each group and plot the line chart of both groups' data on the same graph to compare the difference in scores between the two groups and draw a conclusion.

Finally, we plan to use the Fisher's exact test to ensure that the data are reliable and that the scores meet our expectations because the Fisher's exact test is a test method used for categorical data, helping us determine whether there is a statistical association between two categorical variables. If the p-value is small, we can reject the null hypothesis, indicating that there is a significant association between the two variables.

Fisher's exact test formula:

$$p = \frac{\binom{a+b}{a} \binom{c+d}{c}}{\binom{n}{a+c}} \quad (1)$$

where:

- $a, b, c,$ and d are the observed frequencies in the contingency table,
- $n = a + b + c + d$ is the total sample size,
- $\binom{x}{y} = \frac{x!}{y!(x-y)!}$ is the binomial coefficient.

User feedback

The next step was to analyze the user feedback. We collect data from users by asking them to complete a questionnaire before and after the experiment.

Here is a link to the questionnaire that we used before the experiment:

[Pre-test Questionnaire](#)

This is the link to the questionnaire we used at the end of the experiment:

[Post-test Questionnaire](#)

After getting the data and exporting it, we first pre-process the data. This included removing unnecessary columns and dealing with missing values. We remove irrelevant columns representing timestamps and remove blank rows where the data were NA. The data were then converted to discrete variables of type factor for categorical graphing. Then, we used pie charts and bar charts as visualization methods to display the feedback results.

In particular, the pie chart shows the percentage of responses on whether users would like to participate in the activity again, while the bar chart shows the percentage of distribution on

how much our app can help them understand the concepts of vector addition and subtraction, dot product, fork product, projection, and Magnitude from the users' perspective.

In addition, we have used bar charts to show the percentage distribution of whether users found our APP helpful. These analyses help us visualize user feedback and provide data to support further improvements.

Pioneer Study

We conducted a preliminary user test with three participants: two started by using the APP and one began by reviewing the teaching plan. The participant who reviewed the teaching plan first demonstrated a stronger grasp of theoretical concepts and formulas when answering Test 2. However, this participant also faced challenges, such as difficulty with calculations, the need to use a calculator, and frequent reference to teaching materials to recall key information as the 12 shows.

a) 1

b) 4

c) $p = \frac{P \cdot L}{|P|} = \frac{(1,2,0) \cdot (1,1,1)}{\sqrt{1^2+2^2+0^2}} = \frac{3}{\sqrt{5}}$

d) $j \cdot (k \times l) = j \cdot \begin{bmatrix} 2 \cdot 1 - 2 \cdot 2 \\ 2 \cdot 1 - 2 \cdot 1 \\ 2 \cdot 2 - 2 \cdot 1 \end{bmatrix} = j \cdot \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = -2 \cdot 0 + 2 \cdot 0 = 0$

e) 11

f) Scalar

g) Vector

test 2

Figure 12. Graph of answers of Test 2 from the participant who reviewed the teaching plan first

These issues were mitigated in Test 3, where the app provided guidance, directing users to relevant content quickly and visually representing the data, which helped solve problems and perform accurate calculations. This improvement was evident in the participant's performance in Test 3 as the 13 shows.

In contrast, the participants who used the app first were quicker in answering the questions in Test 2, arriving at the solutions

a) $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \times \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \cdot 2 - 3 \cdot 0 \\ 3 \cdot 2 - 1 \cdot 2 \\ 1 \cdot 0 - 2 \cdot 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ -4 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ -3 \end{bmatrix}$

b) $a \cdot b = 6 + 0 + 2 = 8$
 $a \cdot a = 9 + 0 + 1 = 10$
 $\text{proj} = \frac{a \cdot b}{a \cdot a} \cdot \frac{a}{|a|} = \frac{8}{10} \cdot \frac{1}{\sqrt{10}} \cdot \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} = \frac{4}{5\sqrt{10}} \cdot \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 8/5\sqrt{10} \\ 0 \\ 8/5\sqrt{10} \end{bmatrix}$

c) 1

test 3

Figure 13. Graph of answers of Test 3 from the participant who reviewed the teaching plan first

directly without showing any intermediate steps, as evidenced by their answer sheets, which contained only the final answers. These participants finished early and did not use the entire 10 minutes as the 14 shows.

Upon investigation, it was revealed that the content they missed was not covered by the app. However, the participant who reviewed the teaching plan first was able to answer those questions, highlighting that the teaching plan covered a broader range of topics than the app.

The app, when paired with the teaching plan, allowed participants to answer Test 3 more quickly and accurately. This indicates that traditional instructional materials, when used in conjunction with the app, provide a more comprehensive learning experience. Furthermore, we observed significant improvements in participants' ability to solve complex problems, especially spatial reasoning tasks, when using the app. The visualizations of mathematical concepts of the app, along with its ability to perform accurate calculations, surpassed manual calculations, as some participants admitted to making computational errors that led to incorrect answers.

In terms of these three results, our app proved to be an effective tool in enhancing the learning experience, particularly

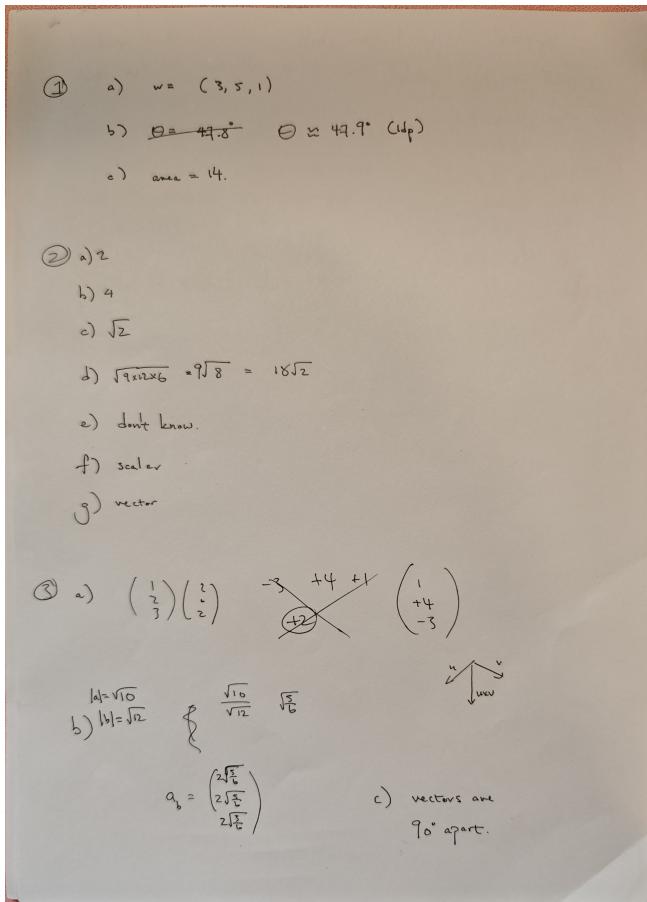


Figure 14. Graph of answers of Test 2 from the participant who used the app first

when used in conjunction with traditional teaching methods. This combination allows students to benefit from both the breadth of knowledge offered by the teaching plan and the app's user-friendly features, such as visual aids and automatic calculations, leading to increased learning efficiency.

Final Results

We analyzed the final data we got from the pre-test questionnaire first.

What is your gender?
16 responses

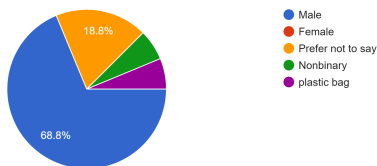


Figure 15. Graph of gender distribution of users

We got a gender distribution plot 15 that shows the majority of respondents (68.8%) identified as male, indicating a predominantly male sample group. Additionally, 18.8% of participants

chose "Prefer not to say," suggesting some respondents preferred to keep their gender private. Smaller portions identified as nonbinary, while one respondent humorously selected "plastic bag."

Do you have any visual impairments that may affect your ability to use AR applications?
16 responses

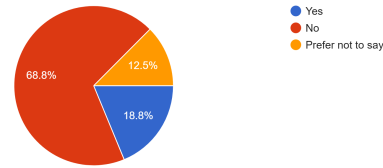


Figure 16. Graph of Visual Impairments Impacting AR Usage of users

A plot of Visual Impairments Impacting AR Usage 16 shows most participants (68.8%) reported not having any visual impairments that would affect their use of AR applications. However, 18.8% indicated that they do have such impairments, while 12.5% preferred not to disclose this information, reflecting a range of visual accessibility considerations among respondents.

What is the highest level of mathematics you have studied?
16 responses

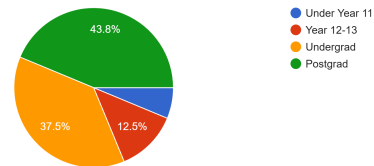


Figure 17. Graph of Mathematics Education Level of users

A plot of Mathematics Education Level shows the survey data 17 shows that the largest segment (43.8%) of respondents have completed postgraduate-level mathematics, suggesting a highly educated sample in terms of math. Meanwhile, 37.5% have studied up to the undergraduate level, and smaller portions had mathematics education at the Year 12-13 or below Year 11 levels.

How easy do you find it to visualise 3D shapes and navigate in 3D environments (including real life)?
16 responses

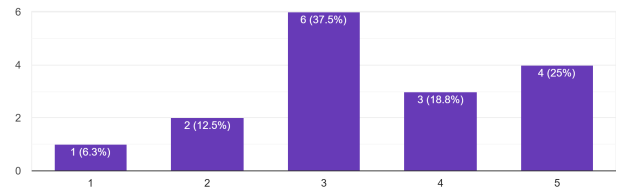


Figure 18. Graph of Ease of Visualizing 3D Shapes and Navigating 3D Environments of users

A plot of Ease of Visualizing 3D Shapes and Navigating 3D Environments 18 shows responses indicate a varied ability to

visualize 3D shapes, with most participants (37.5%) rating their ability as average (3 out of 5). A quarter of the respondents (25%) found it very easy (rating of 5), while fewer participants rated it as difficult (1 or 2), suggesting mixed comfort levels with 3D visualization.

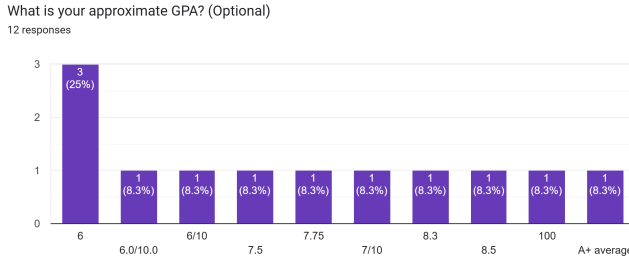


Figure 19. Graph of Approximate GPA of users

A plot of Approximate GPA 19 shows the GPA distribution is spread across a wide range, with the most common GPA being 6.0 out of 10 (25% of respondents). Other GPAs range from 6.0 to a perfect score of 10, with diverse academic performance levels indicated by the responses.

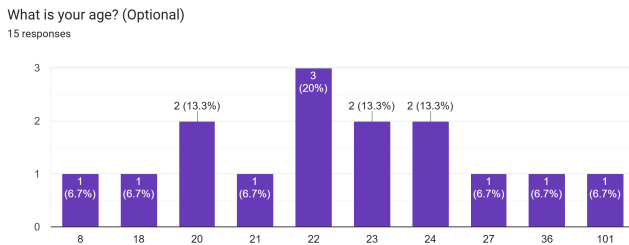


Figure 20. Graph of Age Distribution of users

A plot of Age Distribution 20 shows the age group with the highest representation is 22 years old (20%), but the responses span a wide age range, from as young as 8 to as old as 101, which may be filled by the troublemakers to jam the real data. This indicates a very diverse audience, likely encompassing different stages of education and life experiences.

Next, are the results obtained from the test data. In the form of study, the two control groups were tested for learning results, and the scoring table was obtained according to the significance of the above data to obtain the raw data.

According to the Test 1, the user's individual score is classified and drawn, with a certain degree of fluctuation. 21 Generally speaking, the score of the first question is high while the score of the last question is low. We believe that this may be related to the user's mathematical basis. Although there is a certain degree of understanding difference in the operation of multidimensional vector, it is relatively clear for the operation of vector addition and subtraction, and there are obvious differences in the understanding of cross product.

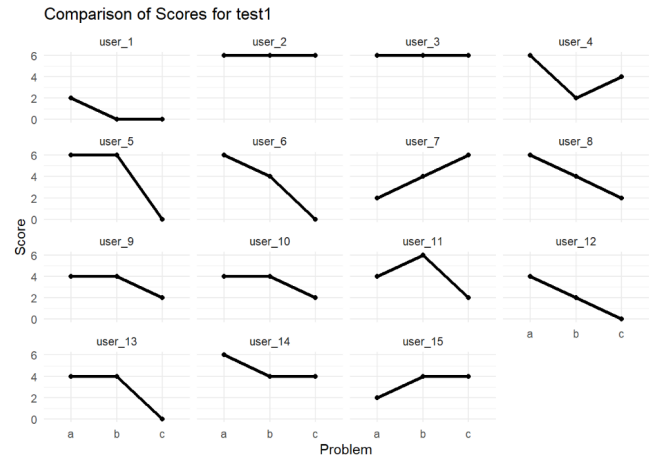


Figure 21. Graph of scores of users on Test 1

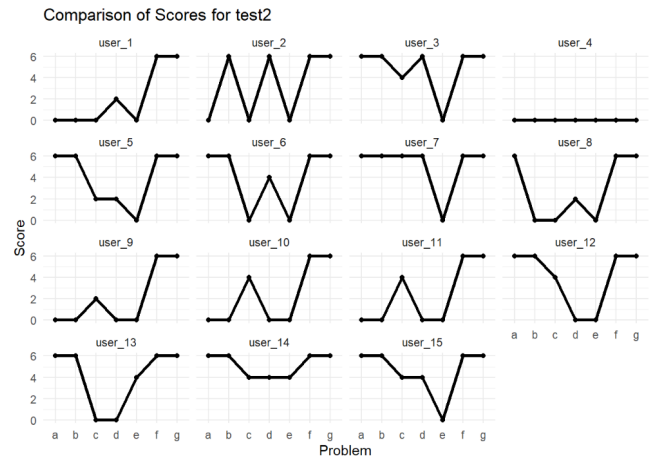


Figure 22. Graph of scores of users on Test 2

In Test 2, the same design problems for different types of concepts mainly focus on cross product and distance. 22 Although we also mentioned dot product, the understanding of this concept that can be determined in Test 1 is relatively clear, and higher scores are obtained in the tests. Next, we compared question a and b (cross product), where the score of the lesson plan was significantly lower than that of users using app. During the interview, we learned that app users could successfully obtain the answer by comparison, but they still had doubts about the principle of the problem, while users with slide showed no score on the concept. Students utilizing the app consistently attained high scores, reaching up to 85.71% correctness, in contrast to 31.11% on average for those employing traditional slides.

Test 3 is divided into three questions, including cross product, projection and dot product. 23 Overall observation, it can be seen that the score still fluctuate to a certain extent, but it is a very obvious polarization. The score of high score users in each question is relatively higher, while the low score users almost show a state that cannot understand the concept. This has some correlation with our test results in Test 2. We

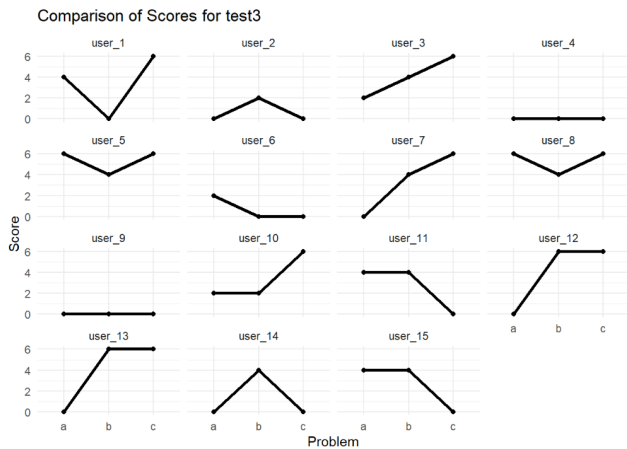


Figure 23. Graph of scores of users on Test 3

believe that it is very difficult for users who are not good at understanding abstract concepts.

During the testing process, we had two control groups. A comparative analysis of the means of the different groups was followed.



Figure 24. Graph of comparison of average scores of users on Test 1

As can be seen from the figure 24, the blue discount is significantly higher than the red broken line, that is, the mathematical foundation of Group 1 (use app first) is overall higher than that of Group 2 (use slides first). Overall, Group 1 scored higher, starting with an average of around 6 on problem 'a' and dropping sharply to near 0 on problem 'c.' In contrast, Group 2 began with a lower average of about 3 on problem 'a' and gradually declined to around 0.5 on problem 'c.' The significant difference at problem 'a' suggests that "App First Users" had a performance lead of roughly 100%, which decreased across the problems. By problem 'c,' both groups' scores converged, indicating a performance gap reduction to near 0%, highlighting a uniform challenge or leveling effect by the last problem.

In Test 2, Group 1 maintained a steady score of about 4 across problems 'a' and 'b,' while Group 2 had consistently lower



Figure 25. Graph of comparison of average scores of users on Test 2

scores, around 2, showing a 50% difference. 25 However, the performance of Group 2 improved significantly on problem 'c,' overtaking Group 1 briefly. Both groups showed variability from 'c' to 'e,' with Group 1 reaching a low of 0 at 'e,' where Group 2 also scored 0, making the gap 0%. Towards the last three problems ('f' and 'g'), Group 1 surged to around 6, a 100% improvement over Group 2 who only reached an average of 3 on 'f' and maintained it at 'g.' The plot reflects fluctuating performances, but Group 2 finished with a marked advantage on the later problems. Specifically, the calculated average score for users employing the app initially was 3.58, whereas the average score for students commencing with the slides was 2.27. Notably, the app users enhanced their scores by 57.7% overall compared to students utilizing slides, thereby demonstrating the app's utility for the users.



Figure 26. Graph of comparison of average scores of users on Test 3

The same trend was seen in Test 3. 26 Group 1 started with an average score of around 3.0 on problem 'a,' slightly decreasing to about 2.8 on 'b,' and then increasing sharply to 4.0 on 'c,' showing an overall improvement of about 33% from 'b' to 'c.' Meanwhile, Group 2 maintained a consistent lower score of 1.5 on problems 'a' and 'b' before rising to 2.5 on 'c,'

marking a 67% increase. Despite this improvement, Group 1 consistently outperformed Group 2 across all problems, with average scores that were about 50% higher on 'a,' nearly 90% higher on 'b,' and around 60% higher on 'c.' This suggests that while both groups improved towards problem 'c,' Group 1 maintained a clear advantage throughout, also demonstrating the app's utility for the users.

After the test, we also recovered the feedback from the post-questionnaire and deployed some analysis on the data we got.

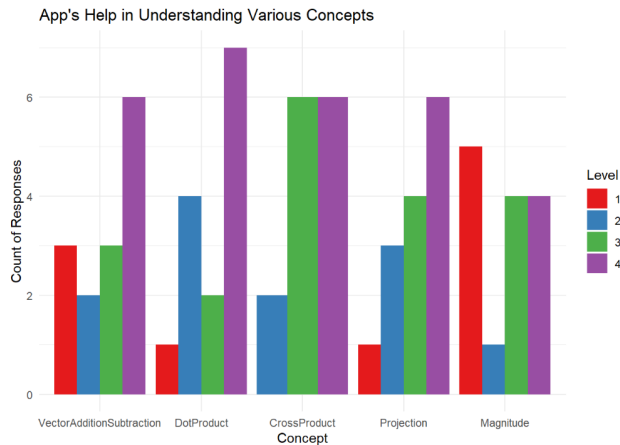


Figure 27. Graph of APP's Help in Understanding Various Concepts for users

The bar chart 27 illustrates how helpful the app was in aiding users' understanding of various mathematical concepts (Vector Addition/Subtraction, Dot Product, Cross Product, Projection, and Magnitude) across four levels (1 to 4), where a higher level indicates greater perceived help. For "Vector Addition/Subtraction," the most responses (4) were at Level 4, indicating strong support, with fewer at lower levels. "Dot Product" saw a more balanced distribution, but with more responses (4) at Level 2, suggesting moderate help. For "Cross Product," Level 4 had the highest count (6), reflecting significant perceived aid from the app. The "Projection" concept responses were highest at Level 3 (5 responses), showing moderate to strong support, while lower levels had fewer responses. Lastly, for "Magnitude," Level 4 (5 responses) and Level 3 (4 responses) were prominent, indicating strong perceived assistance. Overall, our app was generally perceived as helpful across concepts, especially at the higher levels (3 and 4), suggesting that most users felt the app enhanced their understanding effectively, with the majority indicating significant or strong support; only the help in Magnitude is relatively weak.

In terms of willingness analysis 28, the user's willingness to test again is around 50%. We can understand that the dependence on the app or the proportion of potential app users is close to half, which is of great significance for the subsequent improvement of the app function and UI design.

The bar chart 29 provides insights into users' ratings of the app's helpfulness (horizontal axis) and their enjoyment levels (indicated by color within each stacked bar), reflecting patterns of preference and effectiveness for the app in teaching geometry and linear algebra. Most users rated the app as moderately

Willingness to Repeat the Activity

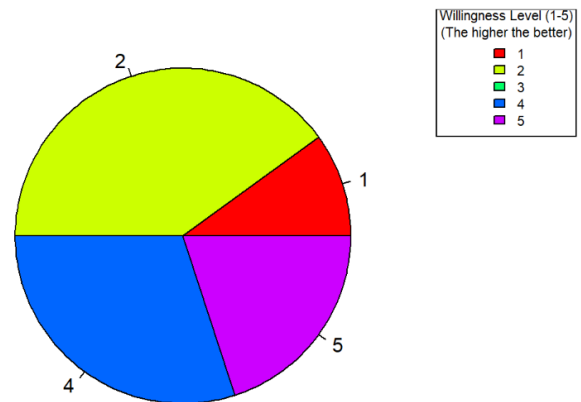


Figure 28. Graph of users' willingness to repeat the activity

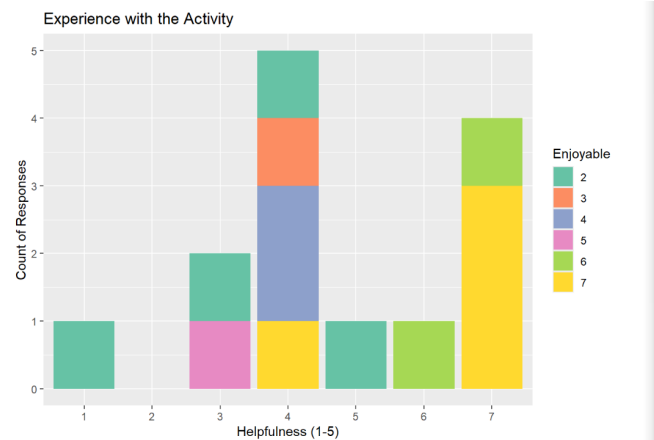


Figure 29. Graph of users' experience with the activity

to highly helpful, with the highest concentration at a score of 4 (5 responses) and another significant group at 7 (4 responses). Those who gave a score of 7 mostly found the activity "very interesting," highlighting a strong positive correlation between high helpfulness and enjoyment. Lower helpfulness scores (1, 2, 3) had fewer responses, suggesting that only a small segment of users found the app less effective. The variation in enjoyment across different helpfulness levels indicates that users' experiences with the app differed, but the consistent trend of higher enjoyment with increased helpfulness suggests that those who benefited more from the app were also more engaged. This pattern underscores the effectiveness of tangible AR in enhancing the learning experience, making complex concepts in geometry and linear algebra easier to understand while also providing a more engaging and enjoyable learning process.

We subsequently performed Fisher's exact test to analyze the correlation of the control group. To satisfy the conditions of Fisher's exact test, we coded the user's score for each question

in each set of tests, specifying 'High' for scores greater than or equal to 4 and 'Low' for scores less than 4. Also, we set the assumptions as follows:

H0: There is no significant difference in the distribution of users' scores on this test

H1: There is significant difference in the distribution of users' scores on this test

```
## Fisher's Exact Test for test1
## [1] "Contingency Table:"
##      ind
## values user_1 user_2 user_3 user_4 user_5 user_6 user_7 user_8 user_9 user_10
## High      0      3      3      2      2      2      2      2      2      2
## Low       3      0      0      1      1      1      1      1      1      1
##      ind
## values user_11 user_12 user_13 user_14 user_15
## High      2      1      2      3      2
## Low       1      2      1      0      1
##
## Fisher's Exact Test for Count Data
##
## data:  contingency_table
## p-value = 0.8028
## alternative hypothesis: two.sided
##
##
## Fisher's Exact Test for test2
## [1] "Contingency Table:"
##      ind
## values user_1 user_2 user_3 user_4 user_5 user_6 user_7 user_8 user_9 user_10
## High      2      4      6      0      4      5      6      3      2      3
## Low       5      3      1      7      3      2      1      4      5      4
##      ind
## values user_11 user_12 user_13 user_14 user_15
## High      3      5      5      7      6
## Low       4      2      2      0      1
##
## Fisher's Exact Test for Count Data
##
## data:  contingency_table
## p-value = 0.004514
## alternative hypothesis: two.sided
##
##
## Fisher's Exact Test for test3
## [1] "Contingency Table:"
##      ind
## values user_1 user_2 user_3 user_4 user_5 user_6 user_7 user_8 user_9 user_10
## High      2      0      2      0      3      0      2      3      0      1
## Low       1      3      1      3      0      3      1      0      3      2
##      ind
## values user_11 user_12 user_13 user_14 user_15
## High      2      2      2      1      2
## Low       1      1      1      2      1
##
## Fisher's Exact Test for Count Data
##
## data:  contingency_table
## p-value = 0.1196
## alternative hypothesis: two.sided
```

Figure 30. Graph of Fisher's exact test results on each Test

The results 30 of Fisher's exact test revealed significant differences in the distribution of user scores across the three tests. In Test 1 and Test 3, the p-values are 0.8028 and 0.1196, respectively, and these values are greater than 0.05, indicating that the differences in the distribution of 'High' (≥ 4) and 'Low' (< 4) scores among different users are not statistically significant. This implies that the differences in user performance in both tests may be random and the null hypothesis that scores are not significantly different between users cannot be rejected. In contrast, the P-value for Test 2 is 0.004514, which is less than 0.05, which indicates that there is a significant difference in the distribution of scores

across users in this test and the null hypothesis can be rejected. This means that in Test 2, the scores of the users who used the app got a significant increase, indicating that our app is helpful in improving the users' maths skills. Meanwhile the results in test 3 are not significant indicating that the users' scores also got improved after using slides to assist them first and then using the APP to assist them, which also proves that the APP is effective.



Figure 31. Graph of word cloud generated from users' feedback on how APP helps you

we analysed the word cloud 31 of open text from the questionnaire regarding user feedback on our app and selected the 10 words with the highest frequency for mapping. The results show that a number of users found our app helpful, as evidenced by the largest percentage of 'yes' words. The most significant results were found for cross product and dot product due to the graph.

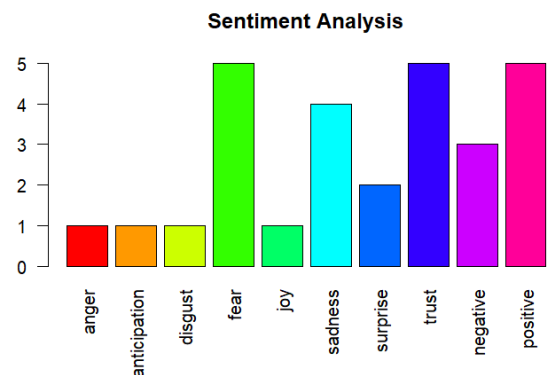


Figure 32. Graph of sentiment analysis based on open-text answers

In addition, we conducted a sentiment analysis 32 of all the open texts covered in the questionnaire. The results show that users' positive sentiment towards the app is 60% higher than negative sentiment, indicating that our app has a positive effect on their understanding of the concepts involved. The very high 'TRUST' score also indicates that our APP is relatively after all reliable. Also high at 5 is 'FEAR', and after examining

the data, we found that users' 'FEAR', 'SADNESS', and negativity were mainly due to the instability of object tracking itself rather than a negative effect on understanding the concept. In addition, a number of users showed surprise with our app, which may indicate that our app is more complete than they thought.

These results generally prove that our app has significantly helped users to improve their learning ability and performance.

LIMITATIONS

Despite these interesting findings, our study has some limitations. Due to the limitations of the software development kit (SDK) that we chose, the application can only recognize specific objects and cannot recognize a wide range of items in a universally applicable way. Another challenge we faced was object recognition interference from the external environment, which sometimes affected the performance of the application. In addition, the small number of participants in this user study limits the statistical significance of our results. More participants are needed to generalize the results. Moreover, there may be some people who are very good at maths, so whether they use the app or not won't affect their grades, so making conclusions that make it difficult to remove that part of the impact won't work. Also, since the data is not normally distributed, tests such as ANOVA cannot be used.

FUTURE WORKS

Our future research will focus on the following points.

1. Finding or developing more stable and adaptable object tracking algorithms, so that the tracking of target objects can be more accurate and fast.
2. Optimise the selection of target objects to make it more universal and at the same time have better tracking effect.
3. further optimise the APP, such as adapting to horizontal screen, shaped screen, etc., so that it is convenient for users to use it in different devices.
4. Conduct a wider range of user studies to draw more statistically significant conclusions.

CONCLUSION

In summary, our research reveals several key insights. First, students often struggle to understand vector addition, dot products, and fork products. Using tangible augmented reality (AR) tools like the ones we used can enhance students' sense of space and understanding of these key concepts. Students with a stronger foundation in mathematics tend to grasp concepts more quickly and use tools more effectively. We also observed that tangible AR tools are particularly beneficial for specific concepts such as the dot product, the fork product, and the point-to-plane distance. Finally, users who were able to accurately and quickly match the corresponding mathematical functions performed better in terms of conceptual understanding. Despite the problems and limitations of our study such as the small number of participants not having a strong statistical significance, and the APP object recognition accuracy not

being high enough or fast enough, in the case of the study at hand, we proved that our APP is effective.

REFERENCES

1. Holton, D. (2005). Foreword. In Findings from the New Zealand Numeracy Development Project 2004 (pp. 1-4). Wellington: Learning Media.
2. Hughes, P., & Peterson, L. (2003). Constructing and using a personal numeracy teaching model in a classroom setting. In L. Bragg, B. C. Campbell, G. Herbert, & J. Mousley (Eds.), *Merino: proceedings of the 26th annual conference of the Mathematics Education Group of Australasia* (Vol. pp. 444–451). Geelong: MERGA.
3. Thomas, G., & Tagg, A. (2005). Evidence for expectations: Findings from the numeracy project longitudinal study. *Findings from the New Zealand Numeracy Development Project 2004*, 21-34.
4. B. Cahyono, M. B. Firdaus, E. Budiman and M. Wati, "Augmented Reality Applied to Geometry Education," *2018 2nd East Indonesia Conference on Computer and Information Technology (EIConCIT)*, Makassar, Indonesia, 2018, pp. 299-303, doi: 10.1109/EIConCIT.2018.8878553.
5. Kaufmann, H., Steinbügl, K., Dünser, A., & Glück, J. (2005). General training of spatial abilities by geometry education in augmented reality. *Annual Review of CyberTherapy and Telemedicine: A Decade of VR*, 3, 65-76.
6. Singh, G., May, Z. (2024). Interactive Learning for Linear Algebra Using Augmented Reality as a Visualisation Approach. In: Romlie, M.F.B., Shaikh Ali, S.H., Hari, Z.B., Leow, M.C. (eds) *Proceedings of the International Conference on Advancing and Redesigning Education 2023. ICARE 2023. Lecture Notes in Educational Technology*. Springer, Singapore.
7. Cerqueira, J., Sylla, C., Moura, J.M., Ferreira, L. (2019). Learning Basic Mathematical Functions with Augmented Reality. In: Brooks, A., Brooks, E., Sylla, C. (eds) *Interactivity, Game Creation, Design, Learning, and Innovation. ArtsIT DLI 2018 2018. Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering*, vol 265. Springer, Cham.
8. Lampropoulos G, Keramopoulos E, Diamantaras K, Evangelidis G. Augmented Reality and Gamification in Education: A Systematic Literature Review of Research, Applications, and Empirical Studies. *Applied Sciences*. 2022; 12(13):6809.
9. Reza Hadi Mogavi, Bingcan Guo, Yuanhao Zhang, Ehsan-Ul Haq, Pan Hui, and Xiaojuan Ma. 2022. When Gamification Spoils Your Learning: A Qualitative Case Study of Gamification Misuse in a Language-Learning App. In *Proceedings of the Ninth ACM Conference on Learning @ Scale (L@S '22)*. Association for Computing Machinery, New York, NY, USA, 175–188.

10. Uriarte-Portillo A, Zatarain-Cabada R, Barrón-Estrada ML, Ibáñez MB, González-Barrón L-M. Intelligent Augmented Reality for Learning Geometry. *Information*. 2023; 14(4):245.
11. Li S, Shen Y, Jiao X, Cai S. Using Augmented Reality to Enhance Students' Representational Fluency: The Case of Linear Functions. *Mathematics*. 2022; 10(10):1718.
12. del Cerro Velázquez F, Morales Méndez G. Application in Augmented Reality for Learning Mathematical Functions: A Study for the Development of Spatial Intelligence in Secondary Education Students. *Mathematics*. 2021; 9(4):369.
13. Zeydan, E. Android vs. IOS: a comparative analysis over mobile operator infrastructures based on crowdsourced mobile dataset. *Telecommun. Syst.*. **78**, 405-419 (2021,11), <https://doi.org/10.1007/s11235-021-00820-y>
14. Nagpal, S., Bansal, S., Kumar, M., Mittal, A. & Saluja, K. Augmented Reality: A Comprehensive Review. *Archives Of Computational Methods In Engineering*. **30** (2022,10)
15. Monteiro, K., Vatsal, R., Chulpongsatorn, N., Parnami, A. & Suzuki, R. Teachable Reality: Prototyping Tangible Augmented Reality with Everyday Objects by Leveraging Interactive Machine Teaching. *Proceedings Of The 2023 CHI Conference On Human Factors In Computing Systems*. (2023), <https://doi.org/10.1145/3544548.3581449>
16. VanDieren, M., Moore-Russo, D., Wilsey, J. & Seeburger, P. Students' understanding of vectors and cross products: Results from a series of visualization tasks. *Proceedings Of The 20th Annual Conference On Research On Undergraduate Mathematics Education*. pp. 991-1000 (2017)